

Econometrics I

Lecture 12b: Binary Choice — Logit and Probit

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Many problems we are interested in look at discrete rather than continuous outcomes:

- ▶ Entering a Market/Opening a Store
- ▶ Working or a not
- ▶ Being married or not
- ▶ Exporting to another country or not
- ▶ Going to college or not
- ▶ Smoking or not
- ▶ etc.

Simplest Example: Flipping a Coin

Suppose we flip a coin which yields heads ($Y = 1$) and tails ($Y = 0$). We want to estimate the probability p of heads:

$$Y_i = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } 1 - p \end{cases}$$

We see some data $Y_1, \dots, Y_N$ which are (i.i.d.)

We know that $Y_i \sim \text{Bernoulli}(p)$.

Simplest Example: Flipping a Coin

We can write the likelihood of N Bernoulli trials as a Binomial:

$$\begin{aligned}\mathbb{P}(Y_1 = y_1, Y_2 = y_2, \dots, Y_N = y_N) &= f(y_1, y_2, \dots, y_N | p) \\ &= \prod_{i=1}^N p^{y_i} (1-p)^{1-y_i} \\ &= p^{\sum_{i=1}^N y_i} (1-p)^{N - \sum_{i=1}^N y_i}\end{aligned}$$

And then take logs to get the **log likelihood**:

$$\ln f(y_1, y_2, \dots, y_N | p) = \left(\sum_{i=1}^N y_i \right) \ln p + \left(N - \sum_{i=1}^N y_i \right) \ln (1-p)$$

Simplest Example: Flipping a Coin

Differentiate the log-likelihood to find the maximum:

$$\begin{aligned}\ln f(y_1, y_2, \dots, y_N | p) &= \left(\sum_{i=1}^N y_i \right) \ln p + \left(N - \sum_{i=1}^N y_i \right) \ln(1 - p) \\ \rightarrow 0 &= \frac{1}{\hat{p}} \left(\sum_{i=1}^N y_i \right) + \frac{-1}{1 - \hat{p}} \left(N - \sum_{i=1}^N y_i \right) \\ \frac{\hat{p}}{1 - \hat{p}} &= \frac{\sum_{i=1}^N y_i}{N - \sum_{i=1}^N y_i} = \frac{\bar{Y}}{1 - \bar{Y}} \\ \hat{p}^{MLE} &= \bar{Y}\end{aligned}$$

That was a lot of work to get the obvious answer: **fraction of heads**.

More Complicated Example: Adding Covariates

We probably are interested in more complicated cases where p is not the same for all observations but rather $p(X)$ depends on some covariates. Here is an example from the Boston HMDA Dataset:

- ▶ 2380 observations from 1990 in the greater Boston area.
- ▶ Data on: individual Characteristics, Property Characteristics, Loan Denial/Acceptance (1/0).
- ▶ Mortgage Application process circa 1990-1991:
 - Go to bank
 - Fill out an application (personal+financial info)
 - Meet with loan officer
 - Loan officer makes decision
 - Legally in race blind way (discrimination is illegal but rampant)
 - Wants to maximize profits (ie: loan to people who don't end up defaulting!)

Financial Variables:

- ▶ P/I ratio
- ▶ housing expense to income ratio
- ▶ loan-to-value ratio
- ▶ personal credit history (FICO score, etc.)
- ▶ Probably some nonlinearity:
 - Very high $LTV > 80\%$ or $> 95\%$ is a bad sign (strategic defaults?)
 - Credit Score Thresholds

Goal $\mathbb{P}(\textit{Deny} = 1 | \textit{black}, X)$

- ▶ Lots of potential **omitted variables** which are correlated with race
 - Wealth, type of employment
 - family status
 - credit history
 - zip code of property
- ▶ Many **redlining** cases hinge on whether or not black applicants were treated in a discriminatory way.

TABLE 11.1 Variables Included in Regression Models of Mortgage Decisions

Variable	Definition	Sample Average
<i>Financial Variables</i>		
<i>P/I ratio</i>	Ratio of total monthly debt payments to total monthly income	0.331
<i>housing expense-to-income ratio</i>	Ratio of monthly housing expenses to total monthly income	0.255
<i>loan-to-value ratio</i>	Ratio of size of loan to assessed value of property	0.738
<i>consumer credit score</i>	1 if no "slow" payments or delinquencies 2 if one or two slow payments or delinquencies 3 if more than two slow payments 4 if insufficient credit history for determination 5 if delinquent credit history with payments 60 days overdue 6 if delinquent credit history with payments 90 days overdue	2.1
<i>mortgage credit score</i>	1 if no late mortgage payments 2 if no mortgage payment history 3 if one or two late mortgage payments 4 if more than two late mortgage payments	1.7
<i>public bad credit record</i>	1 if any public record of credit problems (bankruptcy, charge-offs, collection actions) 0 otherwise	0.074
<i>Additional Applicant Characteristics</i>		
<i>denied mortgage insurance</i>	1 if applicant applied for mortgage insurance and was denied, 0 otherwise	0.020
<i>self-employed</i>	1 if self-employed, 0 otherwise	0.116
<i>single</i>	1 if applicant reported being single, 0 otherwise	0.393
<i>high school diploma</i>	1 if applicant graduated from high school, 0 otherwise	0.984
<i>unemployment rate</i>	1989 Massachusetts unemployment rate in the applicant's industry	3.8
<i>condominium</i>	1 if unit is a condominium, 0 otherwise	0.288
<i>black</i>	1 if applicant is black, 0 if white	0.142
<i>deny</i>	1 if mortgage application denied, 0 otherwise	0.120

First thing we might try is OLS

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- ▶ What does β_1 mean when Y is binary? Is $\beta_1 = \frac{\Delta Y}{\Delta X}$?
- ▶ What does the line $\beta_0 + \beta_1 X$ when Y is binary?
- ▶ What does the predicted value $\hat{Y}$ mean when Y is binary? Does $\hat{Y} = 0.26$ mean that someone gets approved or denied for a loan?

Linear Probability Model

OLS is called the **linear probability model**

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

because:

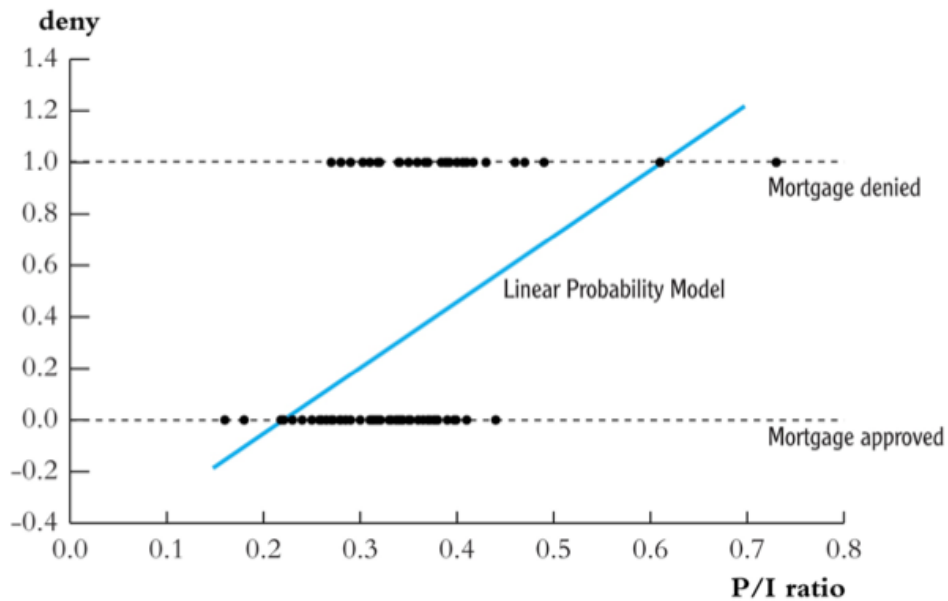
$$\mathbb{E}[Y|X] = 1 \cdot \mathbb{P}(Y = 1|X) + 0 \cdot \mathbb{P}(Y = 0|X)$$

$$\mathbb{P}(Y = 1|X) = \beta_0 + \beta_1 X_i + \varepsilon_i$$

The predicted value is a **probability** and

$$\beta_1 = \frac{\mathbb{P}(Y = 1|X = x + \Delta x) - \mathbb{P}(Y = 1|X = x)}{\Delta x}$$

So β_1 represents the average change in probability that $Y = 1$ for a unit change in X .



- ▶ Is the marginal effect β_1 actually constant or does it depend on X ?
- ▶ Sometimes we predict $\hat{Y} > 1$ or $\hat{Y} < 0$. What does that even mean? Is it still a probability?
- ▶ Fit in the middle seems not so great – what does $\hat{Y} = 0.5$ mean?

Problem with the LPM/OLS is that it requires that **marginal effects are constant** or that probability can be written as linear function of parameters.

$$\mathbb{P}(Y = 1|X) = \beta_0 + \beta_1 X + \varepsilon$$

Some desirable properties:

- ▶ Can we restrict our predictions to $[0, 1]$?
- ▶ Can we preserve **monotonicity** so that $\mathbb{P}(Y = 1|X)$ is increasing in X for $\beta_1 > 0$?
- ▶ Some other properties (continuity, etc.)
- ▶ Want a function $F(z) : (-\infty, \infty) \rightarrow [0, 1]$.
- ▶ What function will work?

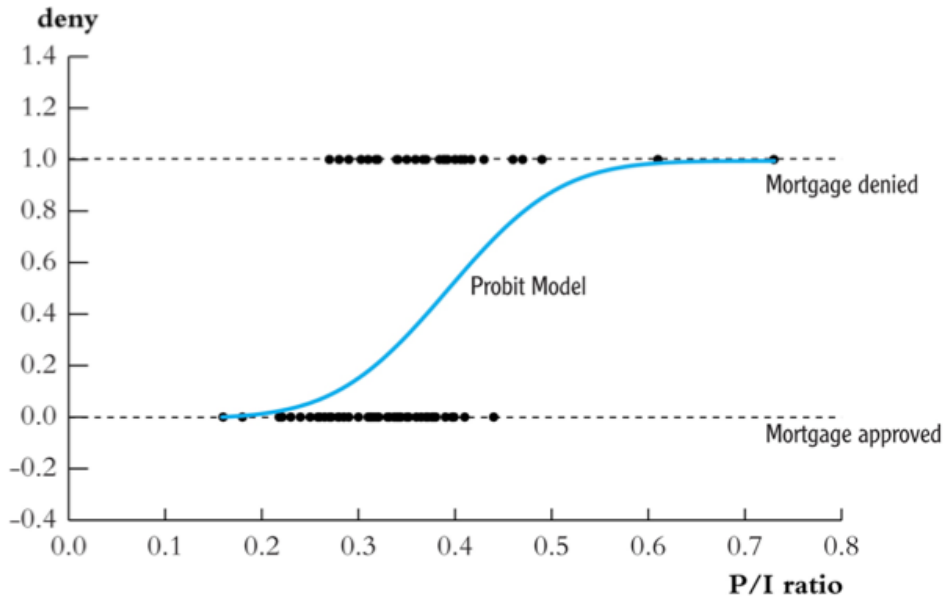
Moving Away from LPM

Problem with the LPM/OLS is that it requires that **marginal effects are constant** or that probability can be written as linear function of parameters.

$$\mathbb{P}(Y = 1|X) = \beta_0 + \beta_1 X + \varepsilon$$

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$$\mathbb{P}(Y = 1|X) = F(\beta_0 + \beta_1 X)$$

- ▶ One $F(\cdot)$ that works is $\Phi(z)$ the normal CDF. This is the **probit** model.
 - Actually any CDF would work but the normal is convenient.
- ▶ One $F(\cdot)$ that works is $\frac{e^z}{1+e^z} = \frac{1}{1+e^{-z}}$ the logistic function . This is the **logit** model.
- ▶ Both of these give 'S'-shaped curves.
- ▶ The LPM is $F(\cdot)$ is the **identity function** (which doesn't satisfy my $[0, 1]$ property).
- ▶ This $F(\cdot)$ is often called a **link function**. Why?

Why use the normal CDF?

Has some nice properties:

- ▶ Gives us more of the 'S' shape
- ▶ $\mathbb{P}(Y = 1|X)$ is increasing in X if $\beta_1 > 0$.
- ▶ $\mathbb{P}(Y = 1|X) \in [0, 1]$ for all X
- ▶ Easy to use – you can look up or use computer for normal CDF.
- ▶ Relatively straightforward interpretation
 - $Z = \beta_0 + \beta_1 X$ is the z -value.
 - β_1 is the change in the z -value for a change in X_1 .

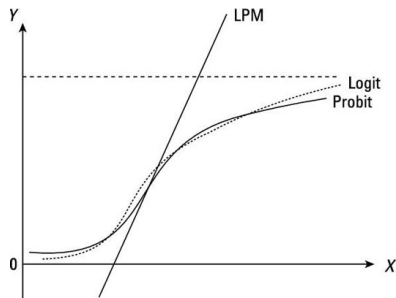
Why use the logistic CDF?

Has some nice properties:

- ▶ Gives us more of the 'S' shape
- ▶ $\mathbb{P}(Y = 1|X)$ is increasing in X if $\beta_1 > 0$.
- ▶ $\mathbb{P}(Y = 1|X) \in [0, 1]$ for all X
- ▶ Easy to compute: $\frac{1}{1+e^{-z}} = \frac{e^z}{1+e^z}$ has analytic derivatives too.
- ▶ Log odds interpretation
 - $\log(\frac{p}{1-p}) = \beta_0 + \beta_1 X$
 - β_1 tells us how **log odds ratio** responds to X .
 - $\frac{p}{1-p} \in (-\infty, \infty)$ which fixes the $[0, 1]$ problem in the other direction.
 - more common in other fields (epidemiology, biostats, etc.).
- ▶ Also has the property that $F(z) = 1 - F(-z)$.
- ▶ Similar to probit but different scale of coefficients
- ▶ Logit/Logistic are sometimes used interchangeably but sometimes mean different things depending on the literature.

A quick comparison

- ▶ LPM prediction departs greatly from CDF long before $[0, 1]$ limits.
- ▶ We get probabilities that are too extreme even for $X\hat{\beta}$ “in bounds”.
- ▶ Some (MHE) argue that though $\hat{Y}$ is flawed, constant marginal effects are still OK.
- ▶ Logit and Probit are highly similar



HMDA: Results

	OLS	Probit	Logit
Constant	-0.174*** (0.026)	-2.96*** (0.205)	-5.56*** (0.406)
Black	0.081*** (0.017)	0.367*** (0.097)	0.657*** (0.177)
Debt/Income	0.471*** (0.087)	2.58*** (0.546)	5.03*** (1.03)
Housing/Income	-0.069 (0.096)	-0.328 (0.652)	-0.405 (1.24)
LTV: Medium	0.028** (0.012)	0.193** (0.081)	0.428*** (0.158)
LTV: High	0.189*** (0.033)	0.779*** (0.175)	1.48*** (0.309)
Consumer Credit	0.031*** (0.004)	0.153*** (0.021)	0.286*** (0.040)
Mortgage Credit	0.019* (0.011)	0.134* (0.074)	0.258* (0.141)
Public Bad Credit	0.200*** (0.023)	0.712*** (0.118)	1.25*** (0.205)
Denied Mortgage Ins	0.701*** (0.041)	2.54*** (0.284)	4.53*** (0.554)
Pseudo R ²	0.51868	0.26407	0.26586
Observations	2,380	2,380	2,380

IID standard-errors in parentheses

*Signif. Codes: *** 0.01 ** 0.05 * 0.1*

- ▶ We cannot compare parameter estimates across specifications
- ▶ Rule of thumb: $\beta_{logit} \approx 1.81 \cdot \beta_{probit}$
- ▶ It is better to compare marginal effects

$$\frac{\partial \mathbb{E}[Y_i=1|X]}{\partial x_k}$$
- ▶ For LPM these are constant β_k , for logit/probit they depend on $X_i\beta$.

We sometimes call these single index models or threshold crossing models

$$Z_i = X_i\beta$$

- ▶ We start with a potentially large number of regressors in X_i but $X_i\beta = Z_i$ is a **scalar**
- ▶ We can just calculate $F(Z_i)$ for Logit or Probit (or some other CDF).
- ▶ Z_i is the **index**. if $Z_i = X_i\beta$ we say it is a **linear index** model.

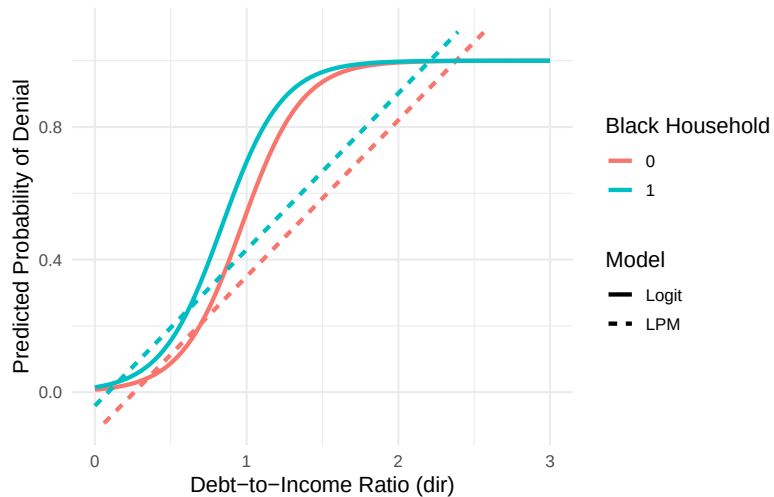
$$\frac{\partial \mathbb{E}[Y_i|X_i]}{\partial X_{ik}} = f(Z_i)\beta_k$$

- ▶ The whole point was that we wanted marginal effects not to be constant
- ▶ So where do we evaluate?
 - Software often plugs in mean or median values for each component
 - Alternatively we can integrate over X and compute:

$$\mathbb{E}_{X_i}[f(Z_i)\beta_k]$$

- The right thing to do is probably to plot the response surface (either probability) or change in probability over all X .

Predicted Loan Denial Probability: LPM vs Logit



An alternative way to think about this problem is that there is a continuously distributed Y^* that we as the econometrician don't observe.

$$Y_i = \begin{cases} 1 & \text{if } Y^* > 0 \\ 0 & \text{if } Y^* \leq 0 \end{cases}$$

- ▶ Instead we only see whether Y^* exceeds some threshold (in this case 0).
- ▶ We can think about Y^* as a **latent variable**.
- ▶ Sometimes you will see this description in the literature, everything else is the same!

- ▶ One temptation might be **nonlinear least squares**:

$$\hat{\beta}^{NLLS} = \arg \min_{\beta} \sum_{i=1}^N (Y_i - \Phi(X_i\beta))^2$$

- ▶ Turns out this isn't what people do.
- ▶ We can't always directly estimate using the log-odds

$$\log \left(\frac{p}{1-p} \right) = \beta X_i + \varepsilon_i$$

- ▶ The problem is that p or $p(X_i)$ isn't really observed.

What does software do?

- Can construct an MLE:

$$\hat{\beta}^{MLE} = \arg \max_{\beta} \prod_{i=1}^N F(Z_i)^{y_i} (1 - F(Z_i))^{1-y_i}$$

$$Z_i = \beta_0 + \beta_1 X_i$$

- Probit: $F(Z_i) = \Phi(Z_i)$ and its derivative (density) $f(Z_i) = \phi(Z_i)$.
Also is **symmetric** so that $1 - F(Z_i) = F(-Z_i)$.
- Logit: $F(Z_i) = \frac{1}{1+e^{-z}}$ and its derivative (density) $f(Z_i) = \frac{e^{-z}}{(1+e^{-z})^2}$ a more convenient property is that $\frac{f(z)}{F(z)} = 1 - F(z)$ this is called the **hazard rate**.

Let $q_i = 2y_i - 1$

$$F(q_i \cdot Z_i) = \begin{cases} F(Z_i) & \text{when } y_i = 1 \\ F(-Z_i) = 1 - F(Z_i) & \text{when } y_i = 0 \end{cases}$$

So that

$$\ell(y_1, \dots, y_n | \beta) = \sum_{i=1}^N \ln F(q_i \cdot Z_i)$$

$$\begin{aligned}\ell(y_1, \dots, y_n | \beta) &= \sum_{i=1}^N y_i \ln F(Z_i) + (1 - y_i) \ln(1 - F(Z_i)) \\ \frac{\partial \ell}{\partial \beta} &= \sum_{i=1}^N \frac{y_i}{F(Z_i)} \frac{dF}{d\beta}(Z_i) - \frac{1 - y_i}{1 - F(Z_i)} \frac{dF}{d\beta}(Z_i) \\ &= \sum_{i=1}^N \frac{y_i \cdot f(Z_i)}{F(Z_i)} \frac{dZ_i}{d\beta} - \sum_{i=1}^N \frac{(1 - y_i) \cdot f(Z_i)}{1 - F(Z_i)} \frac{dZ_i}{d\beta} \\ &= \sum_{i=1}^N \left[\frac{y_i \cdot f(Z_i)}{F(Z_i)} X_i - \frac{(1 - y_i) \cdot f(Z_i)}{1 - F(Z_i)} X_i \right]\end{aligned}$$

FOC of Log-Likelihood (Logit)

This is the **score** of the log-likelihood:

$$\frac{\partial \ell}{\partial \beta} = \nabla_{\beta} \cdot \ell(\mathbf{y}; \beta) = \sum_{i=1}^N \left[y_i \frac{f(Z_i)}{F(Z_i)} - (1 - y_i) \frac{f(Z_i)}{1 - F(Z_i)} \right] \cdot X_i$$

It is technically also a **moment condition**. It is easy for the logit

$$\begin{aligned} \nabla_{\beta} \cdot \ell(\mathbf{y}; \beta) &= \sum_{i=1}^N [y_i(1 - F(Z_i)) - (1 - y_i)F(Z_i)] \cdot X_i \\ &= \sum_{i=1}^N \underbrace{[y_i - F(Z_i)]}_{\varepsilon_i} \cdot X_i \end{aligned}$$

This comes from the hazard rate.

FOC of Log-Likelihood (Probit)

This is the **score** of the log-likelihood:

$$\begin{aligned}\frac{\partial l}{\partial \beta} &= \nabla_{\beta} \cdot \ell(\mathbf{y}; \beta) = \sum_{i=1}^N \left[y_i \frac{f(Z_i)}{F(Z_i)} - (1 - y_i) \frac{f(Z_i)}{1 - F(Z_i)} \right] \cdot X_i \\ &= \sum_{y_i=1} \frac{\phi(Z_i)}{\Phi(Z_i)} X_i + \sum_{y_i=0} \frac{-\phi(Z_i)}{1 - \Phi(Z_i)} X_i\end{aligned}$$

Using the $q_i = 2y_i - 1$ trick

$$\nabla_{\beta} \cdot \ell(\mathbf{y}; \beta) = \sum_{i=1}^N \underbrace{\frac{q_i \phi(q_i Z_i)}{\Phi(Z_i)}}_{\lambda_i} X_i$$

We could also take second derivatives to get the **Hessian** matrix:

$$\begin{aligned}\frac{\partial^2 l}{\partial \beta \partial \beta'} = & - \sum_{i=1}^N y_i \frac{f(Z_i)f(Z_i) - f'(Z_i)F(Z_i)}{F(Z_i)^2} X_i X_i' \\ & + \sum_{i=1}^N (1 - y_i) \frac{f(Z_i)f(Z_i) - f'(Z_i)(1 - F(Z_i))}{(1 - F(Z_i))^2} X_i X_i'\end{aligned}$$

This is a $K \times K$ matrix where K is the dimension of X or β .

The Hessian Matrix (Logit)

For the logit this is even easier (use the simplified logit score):

$$\begin{aligned}\frac{\partial^2 l}{\partial \beta \partial \beta'} &= - \sum_{i=1}^N f(Z_i) X_i X_i' \\ &= - \sum_{i=1}^N F(Z_i)(1 - F(Z_i)) X_i X_i'\end{aligned}$$

This is **negative semi definite**

The Hessian Matrix (Probit)

Recall

$$\nabla_{\beta} \cdot \ell(\mathbf{y}; \beta) = \sum_{i=1}^N \underbrace{\frac{q_i \phi(q_i Z_i)}{\Phi(Z_i)}}_{\lambda_i} X_i$$

Take another derivative and recall $\phi'(z_i) = -z_i \phi(z_i)$

$$\begin{aligned} \nabla_{\beta}^2 \cdot \ell(\mathbf{y}; \beta) &= \sum_{i=1}^N \frac{q_i \phi'(q_i Z_i) \Phi(z_i) - q_i \phi(z_i)^2}{\Phi(z_i)^2} X_i X_i' \\ &= -\lambda_i (z_i + \lambda_i) \cdot X_i X_i' \end{aligned}$$

Hard to show but this is **negative definite** too.

- ▶ If we have the Hessian Matrix, inference is straightforward.
- ▶ $\mathbf{H}_f(\hat{\beta}^{MLE})$ tells us about the **curvature** of the log-likelihood around the maximum.
 - Function is flat $\rightarrow$ not very precise estimates of parameters
 - Function is steep $\rightarrow$ precise estimates of parameters
- ▶ Construct **Fisher Information** $I(\hat{\beta}^{MLE}) = -\mathbb{E}[\mathbf{H}_f(\hat{\beta}^{MLE})]$ where expectation is over the data.
 - Logit does not depend on y_i so $\mathbb{E}[\mathbf{H}_f(\hat{\beta}^{MLE})] = \mathbf{H}_f(\hat{\beta}^{MLE})$.
 - Probit does depend on y_i so $\mathbb{E}[\mathbf{H}_f(\hat{\beta}^{MLE})] \neq \mathbf{H}_f(\hat{\beta}^{MLE})$.
- ▶ Inverse Fisher information $-\mathbb{E}[\mathbf{H}_f(\hat{\beta}^{MLE})]^{-1}$ is an estimate of the variance covariance matrix for $\hat{\beta}$.
- ▶ $\sqrt{\text{diag}[\mathbb{E}[-\mathbf{H}_f(\hat{\beta}^{MLE})]^{-1}]}$ is an estimate for $SE(\hat{\beta})$.

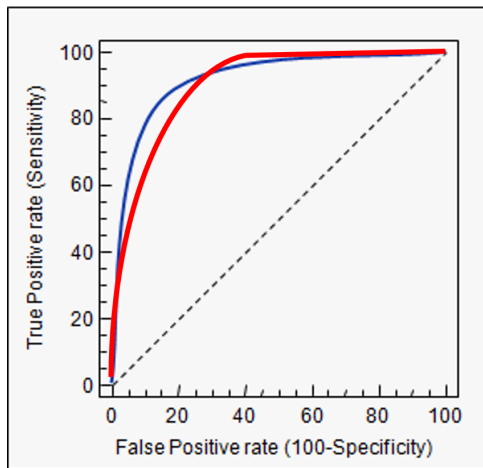
How well does the model fit the data?

- ▶ No R^2 measure (why not?).
- ▶ Well we have likelihood units so average likelihood tells us something but is hard to interpret.
- ▶ $\rho = 1 - \frac{\ell(\hat{\beta}^{MLE})}{\ell(\beta_0)}$ where $\ell(\beta_0)$ is the likelihood of a model with just a constant (unconditional probability of success).
 - If we don't do any better than unconditional mean then $\rho = 0$.
 - Won't ever get all of the way to $\rho = 1$.

Goodness of Fit #2: Confusion Matrix

- ▶ Machine learning likes to think about this problem more like **classification** than regression.
- ▶ A caution: these are **regression** models not **classification** models.
- ▶ Predict either $\hat{y}_i = 1$ or $\hat{y}_i = 0$ for each observation.
- ▶ Predict $\hat{y}_i = 1$ if $\mathbb{P}(y_i = 1 | X_i = x) \geq 0.5$ or $F(X_i\hat{\beta}) > 0.5$.
- ▶ Imagine for cells Prediction: $\{Success, Failure\}$, Outcome $\{Success, Failure\}$
- ▶ Can construct this using the R package caret and command caret.

		Actual Values	
		Positive (1)	Negative (0)
Predicted Values	Positive (1)	TP	FP
	Negative (0)	FN	TN



- At each predicted probability calculate both **True Positive Rate** and **False Positive Rate**.
- AOC is area under the curve

Thanks!
